Alexandria Engineering Journal 119 (2025) 465–477

Contents lists available at ScienceDirect

Alexandria Engineering Journal

journal homepage: www.elsevier.com/locate/aej

Original article

Application of IoT and blockchain technology in the integration of innovation and industrial chains in high-tech manufacturing

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ARTICLE INFO

Keywords:
Industrial ioT
Anomaly detection
Blockchain technology
Predictive maintenance
Smart manufacturing
Real-time ioT monitoring
Secure ioT data management
Data quality

ABSTRACT

In industrial IoT (Internet of Things) environments, accurate anomaly detection and high-quality data management are crucial yet challenging due to noisy and incomplete sensor data. This study introduces BD-IoTQNet, a novel framework designed to address these challenges by integrating data fusion, anomaly detection using the Isolation Forest algorithm, and blockchain-enabled DQM (Data Quality Management). The framework leverages blockchain technology to ensure data transparency and security, while smart contracts automate exception handling to enhance efficiency. Experiments conducted on the NASA Turbofan Engine Degradation and UCI Hydraulic Systems datasets demonstrate that BD-IoTQNet outperforms existing models in accuracy, precision, and data quality improvement, with reduced latency and enhanced robustness under noisy and missing data conditions. An ablation study validates the critical role of each component, showing significant performance drops when modules like DQM or blockchain are excluded. These findings highlight BD-IoTQNet as an effective solution for improving anomaly detection, predictive maintenance, and operational efficiency in industrial IoT systems.

1. Introduction

In the rapidly evolving landscape of the IIoT (Industrial Internet of Things), the integration of vast networks of sensors and devices enables real-time monitoring and predictive maintenance across industrial operations. However, these systems face persistent challenges in managing the accuracy and quality of sensor data due to issues such as noise, incompleteness, and inconsistency. High data volumes, diverse sensor types, and the need for synchronized streams exacerbate these challenges, often resulting in reduced anomaly detection accuracy and operational inefficiencies [1,2].

Traditional predictive maintenance systems struggle to deliver consistent and meaningful insights from heterogeneous sensor data, particularly under real-time conditions. The inability to manage these challenges effectively limits their potential to ensure the reliability and efficiency of industrial systems. Addressing these limitations requires a framework that can seamlessly integrate data from diverse sources, ensure high-quality data streams, and enable accurate, real-time anomaly detection. This paper introduces BD-IoTQNet, a novel framework designed to meet these demands through the integration of advanced data fusion techniques, blockchain-enabled Data Quality Management (DQM), and Isolation Forest-based anomaly detection. Table 1 provides a summary of these challenges and the innovative solutions proposed by BD-IoTQNet.

Although deep learning brings multiple benefits to a wide range of fields, the application of deep learning has still been confronted with various challenges while solving those problems [3]. First of all, successful model training requires large-sized labeled datasets, which cannot be easily available in many industrial scenarios [4]. Second, although deep learning shows outstanding performance in pattern recognition, explainability and transparency lag behind, making anomaly source tracing and data reliability assurance difficult [5]. Moreover, IoT-sourced data can also be noisy, incomplete, or inconsistent, further worsening the accuracy of predictions. These challenges require a method that integrates predictive strengths of deep learning while maintaining data quality and transparency of decisions.

To tackle these challenges, a framework BD-IoTQNet is put forward, which contains three major modules: a data fusion module, an anomaly detection module by using the isolation forest algorithm, and a DQM module. BD-IoTQNet introduces blockchain into IoT to enhance data integrity and transparency from data collection to decision-making, by integrating advanced data processing techniques [6]. Data coming from heterogeneous inputs provided by multiple IoT devices are integrated using a data fusion module for consistency and integrity. The anomaly detection module, featuring the isolation forest algorithm, detects abnormal conditions in real time. A Continuous DQM module monitors

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https://doi.org/10.1016/j.aej.2025.01.020
Received 3 November 2024; Received in revised form 12 December 2024; Accepted 6 January 2025
Available online 10 February 2025
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Table 1
IoT Data Challenges and Proposed Solutions in BD-IoTQNet. This table highlights the key challenges in IoT data management, such as volume and quality, and summarizes how BD-IoTQNet addresses these with data fusion and blockchain.

Challenge	Description	Proposed Solution
Data Volume	Large volumes of data from multiple sensors	Data Fusion and Blockchain for consistency and transparency
Data Variety	Heterogeneous data from different types of sensors	Data Fusion Module to harmonize diverse data sources
Anomaly Detection	Real-time detection of irregularities	Isolation Forest for real-time anomaly detection
Data Quality	Noisy, incomplete, or inconsistent data	DQM Module for continuous quality assessment
Explainability and Transparency	Difficulty in explaining predictions from deep learning models	Blockchain for traceability and transparency

data quality and gives feedback for maintaining reliable input [7]. This framework, in turn, can enhance dependability features for IoT-based systems within manufacturing by enabling holistic solutions with respect to quality improvements in regards to data quality and precision in anomaly detection.

This article underscores the pressing need for an effective IoT data management system in manufacturing, one that integrates blockchain and machine learning to overcome deep learning's current limitations [8]. The BD-IoTQNet framework is designed to be secure, transparent, and scalable, providing a robust solution to industrial IoT data challenges and ultimately boosting the efficiency and accuracy of decision-making processes in these settings.

Our contributions are summarized as follows:

1. Integration of Blockchain and Data Fusion for IoT: The paper presents a novel approach that combines blockchain technology with data fusion techniques to ensure the integrity, traceability, and consistency of heterogeneous IoT data in real-time manufacturing environments.
2. Isolation Forest-based Real-time Anomaly Detection: By utilizing the Isolation Forest algorithm, this research implements an effective method for real-time anomaly detection, addressing the challenge of identifying irregularities in high-dimensional IoT data streams without requiring extensive labeled datasets.
3. Data Quality Management through Automated Feedback Loops: The proposed model introduces a dynamic data quality management system (DQM) that continuously evaluates and enhances the quality of IoT data, using blockchain-enabled feedback mechanisms to improve data reliability and support decision-making.

The rest of this paper is organized as follows: Section 2 reviews related work, including advancements in IoT data fusion, anomaly detection, and blockchain-enabled data management. Section 3 presents the methodology, detailing the BD-IoTQNet framework architecture, including its data fusion, anomaly detection, and data quality management modules. Section 4 describes the experimental setup, datasets, and evaluation metrics, followed by the experimental results and their analysis in Section 4.3. Finally, Section 5 concludes the paper with key findings and future research directions.

2. Related work

Research in IoT data management for manufacturing has evolved significantly, with an emphasis on data fusion, anomaly detection, and ensuring data quality. This section reviews the key concepts in these areas, current advancements, limitations, and trends that highlight the need for further development. By understanding the existing literature, we can identify the gaps this paper aims to address.

2.1. Iot data fusion techniques

Data fusion involves integrating data from multiple sensors or devices to provide a comprehensive and unified view of a system. In IoT-based manufacturing, data fusion is crucial for synthesizing the diverse data streams generated by sensors, actuators, and other IoT devices [9,10]. Traditional methods such as Kalman filtering and Bayesian approaches have been commonly used for multi-sensor data fusion [11], offering basic capabilities to handle sensor noise and combine data for improved decision-making [12,13]. However, these methods face significant scalability challenges as IoT environments grow in complexity, especially with the high-dimensional, high-frequency data produced by modern manufacturing systems.

Recent research has increasingly turned to machine learning techniques, particularly deep learning models like autoencoders and recurrent neural networks (RNNs), to address the limitations of traditional data fusion methods [14]. These models can learn complex, non-linear relationships in data and handle temporal dependencies, making them well-suited for dynamic IoT environments [15]. However, deep learning-based data fusion approaches often require large amounts of labeled data for training, which is typically unavailable in industrial IoT settings. Moreover, while deep learning improves data fusion accuracy, it lacks transparency [16,17], making it difficult to interpret how the model processes data and reaches conclusions.

Despite advancements in machine learning, ensuring the quality of input data during fusion remains an open challenge. IoT data is often incomplete, noisy, or inconsistent, which can compromise the effectiveness of any fusion algorithm [18,19]. Moreover, deep learning models can also imply computational costs, which in turn hamper their practical use when real-time processing may be essential. New research trends to overcome such issues consider the use of blockchain technology together with data fusion. This helps to ensure traceability and integrity of the data cited by [20]. Blockchain allows a decentralized way of recording and checking the origin of the data and how it changes during the fusion process. This adds a layer of transparency and dependability often absent in deep learning models [21]. The paper at hand extends the above ideas further by incorporating blockchain and data fusion toward better quality and higher trustworthiness of IoT data in manufacturing [22].

2.2. Anomaly detection in IoT systems

Anomaly detection is important in IoT-based manufacturing since it identifies the deviation from normal operating patterns, which can imply equipment failures, inefficiency, and even security breaches [23]. Traditionally, statistical methods such as principal component analysis and support vector machines have been used to detect abnormal behaviors from sensor data [24,25]. While these traditional techniques work fine in small-scale systems where data may be structured [26], they fail in efficiency when dealing with the high dimensionality and unstructured nature of IoT data.

The IoT data has grown so complex that deep learning models have become quite famous [27], namely, anomaly detection using LSTMs and CNNs. These models are very outstanding in finding intricate patterns and tiny anomalies in extensive datasets, sometimes hidden in the temporal and spatial characteristics of the data [28,29]. However, deep learning models require large and labeled datasets, which are usually not available in industrial IoT systems [30]. Besides, these methods are usually computation-intensive; hence, their application in real time could be problematic when time is crucial for immediate responses.

Another limitation with most of the existing amorphous deep learning methods is that they are not interpretable [31]. Although they are able to identify anomalies, interpreting why this anomaly has

occurred and tracing back to specific events or devices remains difficult [32]. This becomes a big problem in manufacturing since the potential for swift diagnosis and resolution of problems becomes compromised [33]. Besides, the deep learning models in many cases fail in incomplete/noisy data, which is usually present in IoT environments.

In this respect, this approach has been lately extended with some unsupervised machine learning techniques, proposing methods such as the Isolation Forest algorithm, which does not claim any need of labeled data and is very effective in detecting anomalies in multidimensional data streams [34]. The name justifies Isolation Forest as it randomly partitions the data into segments and identifies those that are more easily separated [35]. This algorithm really works quite efficiently, thus potentially being very suitable for real-time applications [36]. Besides, anomaly detection systems were integrated with blockchain technology for providing a tamper-proof and transparent log of the detected anomalies as a means of having better traceability and accountability [37]. The present paper employed Isolation Forest along with blockchain for improving the efficiency, transparency, and reliability of the anomaly detection process in IoT-driven manufacturing systems.

2.3. Blockchain for IoT data quality management

Blockchain technology has emerged as a solution for ensuring data integrity, security, and transparency in IoT systems [38]. Its decentralized and immutable ledger makes it an ideal platform for managing the vast amounts of data generated by IoT devices in manufacturing environments. In IoT data quality management, blockchain can verify the source of data, ensure that it has not been tampered with [39,40], and provide a traceable record of all data transactions. This ensures that the data used for decision-making in manufacturing is accurate, consistent, and reliable [41].

Most of the current studies in this field have explored the blockchain application for secure data sharing, access control, and privacy protection in IoT networks [42]. Researchers have proposed blockchain-based frameworks that can automate IoT device interactions using smart contracts and ensure data is secure and all transactions are recorded transparently without tampering within the network [43]. It has also been applied to Blockchain in the management of IoT data provenance, where it would be possible to trace the origin of the data that became progressively vital for root cause analysis in manufacturing.

While blockchain does offer great guarantees in the field of data integrity, it does not inherently improve data quality. IoT-generated data usually suffers from noise [44], incompleteness, and inaccurateness, and recording it on a blockchain does nothing to mend these issues. Furthermore, scalability still is one of the main open challenges for blockchain systems nowadays, especially with regard to applications that have to deal with IoT devices generating huge volumes in real time [45,46]. Since blockchain storage is a lot more expensive and slower compared to traditional databases, this really limits its ability to handle large-scale IoT deployments efficiently [47].

In light of these limitations, the latest approaches are considered hybrid ones: using blockchain together with off-chain storage solutions with the aim of reducing overhead from blockchain while ensuring data integrity [48], such as the InterPlanetary File System. In addition, there is a growing interest in integrating blockchain with DQM systems in order to provide qualified continuous monitoring as well as immediate automated feedback to improve the reliability of data [49]. This paper extends these advances by proposing a blockchain-based DQM module that assesses, in real time, the quality of IoT data and provides feedback so that the data remains accurate, consistent, and reliable throughout the system.

3. Methodology

3.1. Overview of the BD-IoTQNet framework

It proposes a blockchain-driven IoT quality management framework, namely BD-IoTQNet, which is the abbreviation for Blockchain-Driven IoT Quality Network. In this paper, a framework was proposed to address the major challenges associated with IoT data management at manufacturing environments. It contains three important embedded modules that are comprised of the data fusion module, an anomaly detection module through the Isolation Forest algorithm, and the Data Quality Management (DQM) module. These all work together in tandem to accomplish appropriate data fusion, effectively detect anomalies, and continuously enhance the quality of IoT data. Blockchain facilitates transparency, traceability, and integrity of data through its whole life cycle.

The overall architecture of the BD-IoTQNet framework is depicted in Figure X, which highlights the modular design and information flow within the system. The diagram illustrates how IoT devices generate data, which is then preprocessed by edge nodes and passed through the data fusion, anomaly detection, and data quality management modules. The processed results are securely recorded and managed via the blockchain layer, ensuring transparency, integrity, and automated response mechanisms. This visual representation provides a comprehensive overview of how the various components interact to achieve robust anomaly detection and data quality enhancement in industrial IoT environments (see Fig. 1).

3.1.1. Framework design

The BD-IoTQNet framework consists of three layers: IoT Device Layer, Data Processing and Analysis Layer, and Blockchain Layer. Each layer plays a critical role in the overall architecture, ensuring real-time data fusion, anomaly detection, and data quality management, while blockchain technology ensures the trustworthiness of the data and the processes.

- IoT Device Layer: This layer is responsible for collecting data from different IoT devices, such as sensors and actuators, embedded in the environment of manufacturing. Because of the nature of things, all these devices generate real-time data; it ranges from simple sensor readings, which include temperature, pressure, and vibration, to operation logs and equipment status. Heterogeneity of data from this layer makes it very complex from several perspectives on data fusion and assurance of quality. This information is then transmitted from the IoT devices to the Data Processing and Analysis Layer for further processing.
- Data Processing and Analysis Layer: This indeed forms the heart of the framework, where it executes its prime tasks of data fusion, anomaly detection, and data quality management. The Data Fusion Module fuses heterogeneous data streams emanating from the IoT Device Layer into a single, unified, and coherent dataset that can then be analyzed. The Anomaly Detection Module uses an Isolation Forest algorithm to identify outliers or anomalies in real-time, hence identifying malfunctioning equipment or abnormal behaviors much sooner than would otherwise be done. Each DQM Module continuously performs quality checks on the data that comes in and ensures that it is accurate, consistent, and complete before using it for decision-making. Results that come from every module are recorded in the Blockchain Layer to ensure transparency and immutability.
- Blockchain Layer: This layer needs to ensure that a system maintains security, transparency, and integrity. Blockchain technology ensures that all activities related to data transaction, such as data fusion results, anomaly detection outputs, and quality ratings, are saved in a decentralized and tamper-proof ledger, and trace any modification of data or process to the contributor of such. Smart contracts in the DQM module automate processes of feedback and correction, allowing the dynamic response of the system to changes in data quality or detected anomalies.

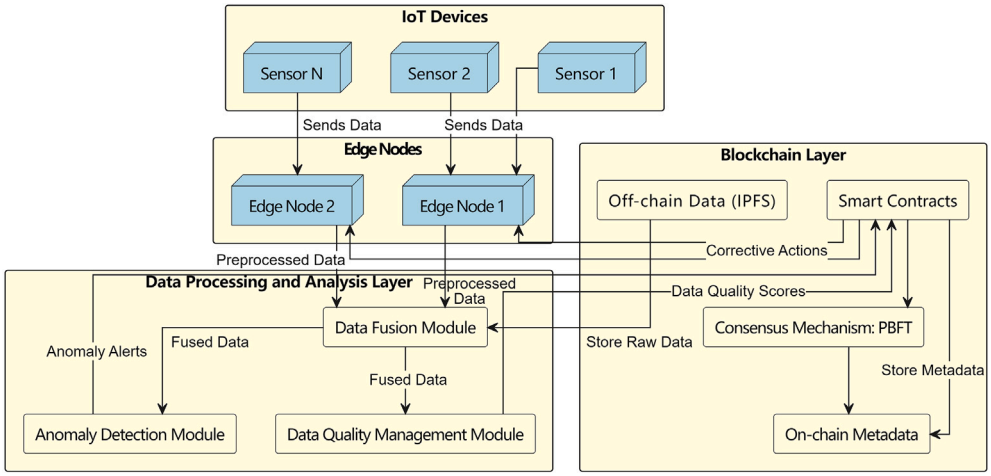


Fig. 1. The overall architecture of the BD-IoTQNet framework, showcasing the modular design and flow of information across IoT devices, edge nodes, data processing modules, and the blockchain layer.

3.1.2. Network architecture

BD-IoTQNet architecture has been designed to operate in a decentralized environment, ensuring robustness, scalability, and efficiency for industrial IoT contexts. The architecture consists of the following key components:

- IoT Nodes: Each IoT device deployed in a manufacturing environment acts as a node, transmitting data to the nearest gateway or directly to the Data Fusion Module. These nodes are responsible for collecting data from sensors and securely forwarding it to the Data Processing and Analysis Layer. The decentralized nature of IoT nodes enhances data collection flexibility and reduces dependency on centralized data sources.
- Edge Nodes: Integrated with an edge computing model, these nodes reduce latency by offloading preliminary computational tasks from central servers. Positioned near IoT devices, edge nodes perform initial anomaly detection, preprocess raw data, and filter out low-value information. This reduces the volume of data that needs to be transmitted and processed by the blockchain network, significantly improving overall system efficiency. By acting as an intermediary, edge nodes optimize data flow and alleviate bottlenecks in real-time data processing.
- Blockchain Network: The blockchain network consists of a series of distributed nodes that collectively maintain a secure and transparent ledger. Each node, hosted by stakeholders such as manufacturers, suppliers, or maintenance teams, stores a copy of the ledger and participates in consensus mechanisms to validate and add transactions, including anomaly detection results or data quality assessments. By decentralizing governance, the blockchain ensures trust and transparency among all parties involved in the manufacturing process.
- Smart Contracts: Smart contracts deployed on the blockchain automate several key functions within the system. For example, they trigger automated actions when an anomaly is detected, such as sending alerts or adjusting production parameters. Smart contracts are also used to initiate corrective actions in the DQM module, ensuring that any data quality issues are addressed immediately without manual intervention. By integrating PBFT (Practical Byzantine Fault Tolerance), the framework maintains robust data integrity and operational reliability while meeting the scalability and performance demands of industrial IoT systems.

BD-IoTQNet incorporates advanced strategies to tackle challenges posed by high-dimensional and complex datasets. In the Data Fusion Module, dimensionality reduction techniques like Principal Component Analysis (PCA) are applied to identify the most critical features, balancing computational efficiency and data fidelity. Furthermore, attention-based dynamic feature selection mechanisms enable the system to adapt to diverse and complex sensor data streams. In the Anomaly Detection Module, the Isolation Forest algorithm is augmented by deep learning models such as Variational Autoencoders (VAEs) to capture intricate patterns in high-dimensional data. These enhancements ensure that the framework maintains high accuracy and scalability, even under demanding conditions.

To further address the computational demands of large-scale datasets, BD-IoTQNet employs distributed computing frameworks and GPU-based parallel processing, which reduce latency and support real-time decision-making. The blockchain layer enhances scalability through sharding techniques, allowing the system to process large datasets efficiently while maintaining security and transparency. Additionally, the DQM Module incorporates advanced imputation methods, such as generative models, to handle noise and missing data, providing reliable input for downstream anomaly detection. Together, these integrated strategies make BD-IoTQNet highly adaptable, robust, and efficient in managing high-dimensional data across diverse industrial scenarios.

3.1.3. Interaction between modules

The interaction between the three modules ensures seamless data flow and coordination. As data is collected from the IoT devices, it is first processed by the Data Fusion Module, which consolidates and cleanses the data. This fused data is then passed through the Anomaly Detection Module, where Isolation Forest analyzes the data for potential outliers. Simultaneously, the DQM Module evaluates the data quality, applying smart contracts to trigger automated feedback if any quality issues are detected. The results from all three modules are recorded on the blockchain, ensuring that all actions are transparent and traceable while maintaining data integrity and security.

While the reliance on blockchain technology introduces computational overhead, BD-IoTQNet employs multiple strategies to optimize performance and reduce resource consumption. The framework adopts a Practical Byzantine Fault Tolerance (PBFT) consensus mechanism, which reduces latency and energy consumption compared to traditional blockchain protocols, ensuring suitability for real-time IoT applications. To further alleviate computational demands, the framework utilizes a hybrid storage approach: high-volume raw sensor data is stored off-chain using systems like the IPFS (InterPlanetary File System), while only critical metadata and quality assessments are maintained on-chain. This design minimizes blockchain's computational and storage burdens while retaining its transparency and security features.

Additionally, BD-IoTQNet incorporates dynamic sharding techniques, dividing the blockchain into smaller, manageable segments that process transactions in parallel. This significantly reduces processing times and ensures scalability as the system handles higher data volumes. Adjustable block sizes and transaction batching further enhance the system's efficiency by enabling flexible data handling based on workload and application requirements.

To further enhance the framework's capability to manage missing and noisy data, the DQM Module can integrate advanced imputation techniques such as Generative Adversarial Networks (GANs) and transformer-based models. GAN-based imputation leverages the generator-discriminator structure to predict missing values while maintaining data fidelity, whereas transformer-based models use attention mechanisms to capture complex temporal dependencies in time-series data. These advanced methods complement the existing imputation strategies, allowing BD-IoTQNet to handle more complex data patterns efficiently. This integration not only improves the quality of the input data but also enhances the overall anomaly detection accuracy and robustness.

In essence, BD-IoTQNet is designed to provide a robust, scalable, and transparent solution to the challenges of IoT data management in manufacturing environments. By integrating blockchain with advanced data processing techniques, including state-of-the-art imputation methods, and optimizing its implementation, the framework achieves a balance between computational efficiency and the critical need for data security and transparency. These enhancements ensure reliable IoT data, efficient anomaly detection, and continuous data quality improvements in industrial settings.

3.2. Data fusion module

In BD-IoTQNet, the data fusion process employs Kalman filters and weighted averaging to consolidate data from heterogeneous IoT sources. This ensures that the final input data used for anomaly detection is both reliable and consistent. The weighted averaging method dynamically assigns weights to each data source based on its reliability and relevance at any given time. These weights are calculated in real time using a combination of statistical measures, such as variance and historical performance of each sensor, as well as contextual factors like environmental conditions and operational status. For instance, sensors with higher variance or frequent deviations from expected values are assigned lower weights, whereas sensors with consistent and accurate readings contribute more heavily to the final fused output.

To enable real-time adjustments, the framework integrates an adaptive weighting mechanism within the Data Fusion Module. This mechanism continuously monitors incoming data streams and recalibrates the weights as new data becomes available. Kalman filters play a critical role in this process by estimating the state of each sensor signal and predicting the reliability of future readings based on observed trends. These predictions inform the weight adjustment process, ensuring that the data fusion remains robust even under dynamically changing conditions, such as sensor noise, drift, or partial failures.

This dynamic weighting and real-time adjustment capability significantly enhances the accuracy and robustness of the data fusion process, ensuring that BD-IoTQNet can provide high-quality input data to downstream modules, such as anomaly detection and data quality management. By prioritizing the most reliable data sources and adapting to real-time conditions, the framework is better equipped to handle the inherent uncertainties and variability of industrial IoT environments.

The data fusion process begins with the standardization of raw sensor data. For each sensor S_i, the data $x_i(t)$ is collected at time t, where the sensors can provide different types of measurements (e.g., temperature, pressure, vibration). The first step is to normalize all sensor data to a common scale:

$$x_i^{\text{norm}}(t) = \frac{x_i(t) - \mu_i}{\sigma_i} \tag{1}$$

where $x_i^{\text{norm}}(t)$ is the normalized value of the sensor data, μ_i is the mean of sensor S_i's data, and σ_i is the standard deviation.

After normalization, the system applies a weighted averaging method to combine the sensor data. The final fused data $y(t)$ is calculated as:

$$y(t) = \sum_{i=1}^{n} w_i \cdot x_i^{\text{norm}}(t) \tag{2}$$

where $y(t)$ is the fused data at time t, w_i represents the weight assigned to sensor S_i, and n is the total number of sensors. The weights w_i are typically based on the reliability of the sensors, which can be determined through historical performance or real-time reliability assessments.

In cases where the data involves time-varying measurements, such as dynamic environmental factors, the Kalman filter is employed to estimate the true value $\hat{x}(t)$ of the underlying process. The Kalman filter can be described by the following recursive equations:

$$\hat{x}(t|t-1) = A \cdot \hat{x}(t-1|t-2) + B \cdot u(t-1) \tag{3}$$

where $\hat{x}(t|t-1)$ is the predicted state estimate at time t given observations up to time $t-1$, A is the state transition matrix, and B is the control input matrix with $u(t-1)$ representing external control inputs.

The predicted value $\hat{x}(t|t-1)$ is then corrected using the measurement $z(t)$ from the sensors, which gives the updated estimate $\hat{x}(t)$:

$$\hat{x}(t) = \hat{x}(t|t-1) + K(t) \cdot (z(t) - H \cdot \hat{x}(t|t-1)) \tag{4}$$

where $K(t)$ is the Kalman gain, H is the measurement matrix, and $z(t)$ is the measurement at time t. The Kalman gain $K(t)$ is computed as:

$$K(t) = P(t|t-1) \cdot H^{\top} \cdot \left(H \cdot P(t|t-1) \cdot H^{\top} + R\right)^{-1} \tag{5}$$

where $P(t|t-1)$ is the predicted error covariance, and R is the measurement noise covariance.

Finally, the error covariance is updated for the next iteration as follows:

$$P(t) = (I - K(t) \cdot H) \cdot P(t|t-1) \tag{6}$$

where $P(t)$ is the updated error covariance, and I is the identity matrix.

The data fusion process described above plays a critical role in modern IoT-based manufacturing environments. In such settings, sensor data is often noisy and subject to uncertainty, requiring robust techniques to ensure that the resulting data used for decision-making is reliable. The advantages of this approach include:

1. Normalization, weighted averaging, and Kalman filtering can improve data accuracy by reducing noisy or incomplete data.
2. Because the Kalman filter is recursive, estimates keep updating and refining in real time. Thus, it becomes applicable in dynamic manufacturing processes.
3. It will be easy to upscale the fusion model in case there is a need to add sensors or data sources. It is also very adaptable in complex manufacturing environments.

Data fusion within the BD-IoTQNet framework plays a key role in ensuring data reliability for further analysis, such as anomaly detection and data quality management. It has to make sure that the heterogeneous streams of data originating from IoT devices are integrated in a consistent manner and under a standardized framework. This process of fusion directly influences the performance of the successive module for anomaly detection, whereby correct and reliable data is fed into it. Secondly, the incorporation of blockchain in this module makes the whole process of fusing data transparent and traceable, wherein the steps taken in the process of fusing data are traceably and immutably recorded on the blockchain. It checks that the output of the fusion process is reliable and any inconsistency in the data quality can be tracked back to its source. The Data Fusion Module thus plays an integral part in ensuring the overall functionality of the BD-IoTQNet system since it ensures correct, consistent high-quality data that guarantees real-time decision-making for manufacturing.

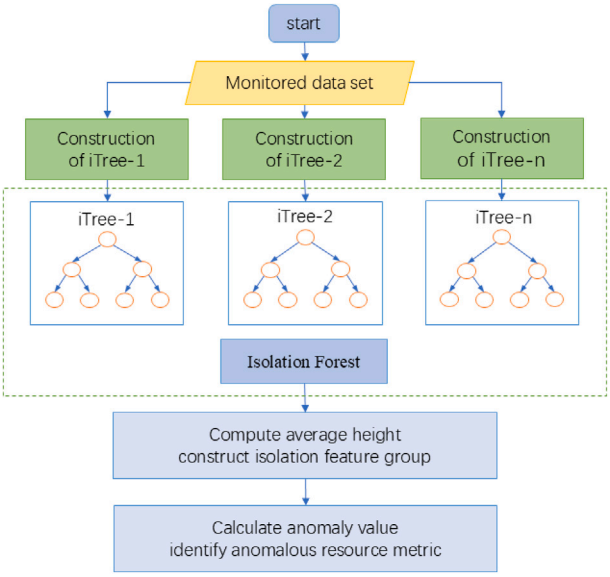


Fig. 2. The structure of the Isolation Forest algorithm. Shows how the Isolation Forest algorithm isolates anomalies using shorter path lengths in random trees".

3.3. Anomaly detection

The core role of the BD-IoTQNet framework revolves around detecting abnormal behavior in IoT-generated data streams through the Anomaly Detection Module-these data streams being all-important in real-time manufacturing environments. Its main purpose is to identify deviations in equipment performance or system behaviors that would allude to faults, inefficiencies, or security issues. The key idea of the module is based on the Isolation Forest algorithm, one of the most common unsupervised machine learning anomaly detection algorithms used in large and high-dimensional data. As this type of model has been chosen, the developed system will be able to predict outliers without labeled data, which are usually quite rare for manufacturing processes. Fig. 2 shows how the Isolation Forest algorithm works.

The Isolation Forest algorithm isolates anomalies by constructing multiple decision trees, where data points that can be separated with fewer splits are more likely to be anomalies. The algorithm's core concept is that anomalies are "isolated" earlier because they exhibit different behaviors compared to the majority of the data. Mathematically, the anomaly score $s(x)$ quantifies the likelihood of a data point x being an anomaly. It is calculated as follows:

$$s(x) = 2^{-\frac{E(h(x))}{c(n)}} \tag{7}$$

Here, $E(h(x))$ represents the expected path length of data point x in the isolation trees, which is calculated as the average path length across all trees in the Isolation Forest. The path length $h(x)$ for a data point refers to the number of edges traversed from the root node to the terminal node (leaf) where the point is isolated. This expectation is crucial as anomalies, being few and different, are isolated with shorter path lengths compared to normal points, which require more splits for isolation.

The algorithm evaluates whether the anomaly score $s(x)$ exceeds a threshold τ, which differentiates normal data points from anomalies. The threshold τ is typically chosen between 0.6 and 0.8 based on specific application requirements:

$$\text{Anomaly}(x) = \begin{cases} \text{True}, & \text{if } s(x) > \tau \\ \text{False}, & \text{if } s(x) \leq \tau \end{cases} \tag{8}$$

where τ is the threshold that differentiates normal points from anomalous ones. Typically, a threshold between 0.6 and 0.8 is chosen to optimize the detection accuracy based on the specific requirements of the industrial system.

Despite its numerous advantages, the Isolation Forest algorithm has certain limitations, particularly its susceptibility to false positives when the data distribution is highly imbalanced or when noise is misclassified as anomalies. To address this, the BD-IoTQNet framework integrates the Data Quality Management (DQM) Module to filter and preprocess input data, significantly reducing the likelihood of noise-triggered false positives. However, further exploration of alternative unsupervised models, such as Density-Based Spatial Clustering of Applications with Noise (DBSCAN), is planned. DBSCAN, for instance, excels in identifying anomalies within dense clusters and non-linear data distributions, making it a valuable comparison point for evaluating the Isolation Forest's effectiveness in varying industrial scenarios.

In practical deployment, the Isolation Forest algorithm's scalability and efficiency make it suitable for the online monitoring of high-dimensional data in complex manufacturing systems. This is particularly critical for identifying previously unseen anomalies in dynamic environments where labeled data is unavailable or expensive to obtain.

Smart contracts on the blockchain make the module even more functional: when an anomaly is detected, a response, already predetermined to range from mere alerts to changes in the operational parameters or even workflows of maintenance, is fired autonomously. Thus, automation of responses reduces the time to act upon the detected anomalies such that potential issues can be fixed before they escalate to larger-sized problems. These two capabilities-real-time detection of anomalies and automated responses-merged provide a fluent, effective way to manage the risks of manufacturing systems.

In the extended scope of BD-IoTQNet, the Anomaly Detection Module closely interacts with the Data Fusion Module and the Data Quality Management Module. Data fusion ensures that sensor data coming from various sources is in a combined, standardized form to alleviate the impact of noisy or missing data on the anomaly detection process. In parallel, the anomaly detection results are valuable feedback for the Data Quality Management Module, and the latter constantly updates the information in its evaluation and enhancement regarding the reliability of incoming data. These modules work together to make sure that data anomalies are detected as early as possible, addressed as fast as possible, and documented transparently.

3.4. Data quality assessment and feedback

In fact, DQM in the BD-IoTQNet is for the purpose of ensuring that the IoT data streams transmitted within a manufacturing environment are reliable, accurate, and consistent. Present industrial systems are based on advanced technology, which means poor-quality data will lead to defective decisions, malfunctioning equipment, and inefficient production. Quality data becomes crucial in those situations. It continuously observes the incoming data and assesses it, with automated feedback mechanisms used to enrich the data in order to enhance its overall quality before making use of it-be it for analysis, anomaly detection, or decision-making.

The core mission of the DQM Module is to make IoT data reliable along the whole data pipeline since IoT data are too often affected by noise, incompleteness, or inconsistencies. This module watches incoming data streams in real time for possible problem points such as missing values, outliers, or sensor failures. Depending on the gravity of a detected issue and the imminent impact of a degradation of data quality, when such issues are detected, the DQM module can automatically correct them or flag them for further review.

The data quality in IoT systems can be quantified and managed by employing several metrics, including accuracy, completeness, timeliness, and consistency. The DQM module continuously evaluates these parameters and uses a combination of historical performance and real-time data monitoring to assess the quality of the incoming data. Mathematically, the quality score $Q_i(t)$ for a data stream from sensor S_i at

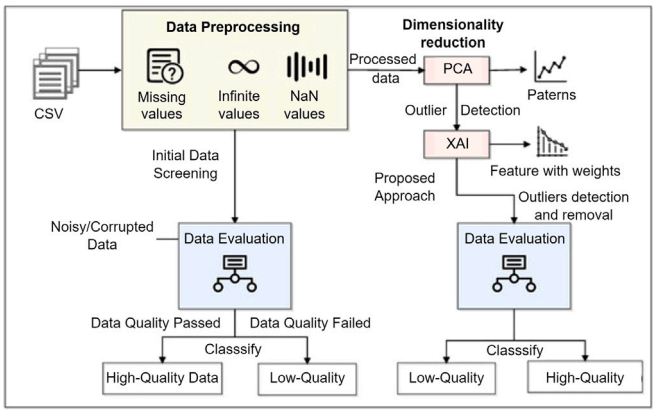


Fig. 3. Data Quality Management (DQM) Workflow for IoT Data. Illustrates the steps of the DQM module, including data preprocessing, quality evaluation, and automated corrections.

time t can be defined as:

$$Q_i(t) = w_1 \cdot A_i(t) + w_2 \cdot C_i(t) + w_3 \cdot T_i(t) + w_4 \cdot I_i(t) \tag{9}$$

where $A_i(t)$ represents the accuracy of the data, $C_i(t)$ is the completeness (i.e., the ratio of available data to expected data), $T_i(t)$ is the timeliness (i.e., whether data is received within the expected time frame), and $I_i(t)$ is the consistency of the data. The weights w_1, w_2, w_3, w_4 are coefficients that reflect the relative importance of each quality factor, which may vary based on the specific application or data type.

If the quality score $Q_i(t)$ falls below a predefined threshold τ_q, the DQM module initiates a corrective action. This process can involve either automated data cleaning or sending alerts to operators, who may need to intervene manually. The condition for triggering a corrective action can be expressed as:

$$\text{CorrectiveAction}(i,t) = \begin{cases} \text{True}, & \text{if } Q_i(t) < \tau_q \\ \text{False}, & \text{if } Q_i(t) \geq \tau_q \end{cases} \tag{10}$$

In the case where corrective actions are triggered, the DQM module leverages historical data and predictive algorithms to either impute missing values or correct noisy data. Imputation techniques can involve simple methods such as mean or median imputation, or more advanced statistical methods depending on the complexity of the data. Fig. 3 provides a detailed architectural diagram of the DQM module's workflow. This figure illustrates the process from identifying data issues (such as missing values or noise) to correcting them, and ultimately recording all actions on the blockchain for traceability.

Because the DQM Module is inextricably linked with Blockchain technology, it ensures that all changes to the data-correcting, imputing, flagging for manual review-are transparently and immutably recorded. Herein, every corrective action that will be enacted by the DQM module shall be logged on the blockchain for an unalterable audit trail. This enables any stakeholder down the chain of manufacturing to observe data life from the sensor level right through to decision-making.

Beyond just solving immediate problems of data quality, the DQM Module is proactive, housing predictive algorithms observing trends in sensor performance over time. For example, if over a number of time periods, the quality score $Q_i(t)$ of a sensor has been showing a downwards trend, then the DQM module will be able to predict likely failures in the near future and schedule preventive maintenance in advance of the sensor's full failure. Predictive capability like this has great value in preventing down-time and ensuring the smooth running of manufacturing processes.

Conclusively, the Data Quality Management Module is an imperative module for BD-IoTQNet to ensure the accuracy, consistency, and dependability of IoT data on which the analysis and decisions are made. The module continuously observes streams, detects several related quality issues, and applies suitable automated corrective actions toward assuring the system's dependability. It has integrated blockchain technology to bring transparency and accountability, while the utilization of smart contracts shows that data quality management processes can be automatic and swift. This proactive approach to data quality, in turn, enhances the performance of the BD-IoTQNet framework to bring overall stability and efficiency in IoT-driven manufacturing environments.

3.5. IoT-blockchain network fusion

The integration of IoT and blockchain networks in BD-IoTQNet is a key innovation designed to enhance data integrity, transparency, and reliability across industrial systems. This fusion is achieved through a seamless interaction between real-time IoT data streams and blockchain's decentralized, tamper-proof infrastructure. When IoT devices generate data, edge nodes act as intermediaries, aggregating real-time sensor readings and preparing them for blockchain storage. Each data transaction is enriched with metadata, including timestamps, sensor IDs, and data quality scores, ensuring that all information recorded on the blockchain is traceable and verifiable.

To address the computational challenges of blockchain in high-volume IoT systems, BD-IoTQNet adopts a hybrid storage strategy. Raw sensor data is stored off-chain in systems like the InterPlanetary File System (IPFS), while only critical metadata and quality assessments are stored on-chain. This reduces storage and computational overhead while maintaining blockchain's transparency and reliability. Furthermore, the blockchain operates on a Practical Byzantine Fault Tolerance (PBFT) consensus mechanism, which ensures high throughput and low latency, making it well-suited for dynamic IoT environments. This consensus mechanism also provides robust security, protecting the network from tampering and malicious attacks.

While PBFT inherently reduces the risk of common blockchain attacks such as 51% attacks, potential vulnerabilities still exist. A 51% attack would require a malicious actor to compromise the majority of the pre-approved validator nodes, which is highly unlikely given the distributed nature of the network and the reliance on multiple voting rounds to reach consensus. To further enhance security, BD-IoTQNet incorporates network monitoring mechanisms to detect anomalous behaviors among validator nodes. Smart contracts are designed to isolate compromised nodes and trigger alerts, ensuring that the blockchain network remains secure and operational. Additionally, the hybrid blockchain architecture minimizes the impact of potential breaches by limiting sensitive data exposure on-chain. These measures collectively ensure a robust and resilient blockchain layer, capable of maintaining data integrity and transparency even in the face of sophisticated threats.

Another critical component of the IoT-blockchain fusion is the use of smart contracts. These contracts automate essential processes, such as triggering anomaly detection when sensor readings exceed predefined thresholds, initiating corrective actions for data quality issues, and maintaining real-time updates of data quality scores. This automation minimizes human intervention, reduces response times, and ensures consistent system behavior. Additionally, the blockchain interacts continuously with the Data Quality Management (DQM) module, forming a feedback loop where real-time quality assessments are logged on the blockchain and used to refine system performance. Through these mechanisms, the IoT-blockchain integration in BD-IoTQNet ensures that industrial systems operate with high data reliability, transparency, and efficiency.

Table 2
NASA Prognostics Data Repository - Turbofan Engine Degradation Simulation Dataset. The dataset provides high-dimensional time series data, ideal for testing anomaly detection and predictive maintenance.

Characteristic	Description
Number of engine units	100 engines, each with unique operational history
Sensors	21 sensors (temperature, pressure, vibration, fuel flow)
Cycles per unit	Varies (each unit is run until failure)
Data format	Time series data recorded at each operational cycle
Features	Operational settings (thrust, altitude) and sensor readings

Table 3
UCI Machine Learning Repository - Condition Monitoring of Hydraulic Systems Dataset. This dataset is well-suited for evaluating data fusion and quality management modules in industrial IoT environments.

Characteristic	Description
Monitored units	Multiple hydraulic systems
Sensors	4 sensors (pressure, volumetric flow, temperature, cooling power)
Time interval	Data recorded every second
Features	System condition labels along with sensor measurements

4. Experiment

4.1. Datasets

To evaluate the performance and efficiency of the BD-IoTQNet framework, two very popular datasets are considered in this work: NASA Prognostics Data Repository - Turbofan Engine Degradation Simulation Dataset and UCI Machine Learning Repository - Condition Monitoring of Hydraulic Systems Dataset. These datasets are well known in the predictive maintenance and industrial IoT community and hence are perfect candidates for the testing of the capabilities of the data fusion, anomaly detection, and data quality management modules within the framework.

NASA Prognostics Data Repository - Turbofan Engine Degradation Simulation Dataset [50]: This dataset simulates the operational data from turbofan engines over their full life cycle, from nominal performance to failure. High-dimensional sensor readings are generated for each unit through operation cycles of the engines. It is of particular interest when evaluating the performance of the BD-IoTQNet framework for anomaly detection and predictive maintenance. Details about this dataset are summarized in the following 2:

Condition Monitoring of Hydraulic Systems Dataset from UCI Machine Learning Repository [51]: The second dataset is from the UCI Machine Learning Repository, addressing condition monitoring for hydraulic systems in heavy machinery. It includes data recorded from several hydraulic units operating and sensor measurements that monitor critical hydraulic parameters such as pressure and temperature. The dataset is of utmost relevance to the framework in terms of data quality management and fusion evaluation. Details are as shown in Table 3:

By using these datasets, the BD-IoTQNet framework capability on handling the real-world industrial data challenge is comprehensively evaluated. The NASA dataset allows comprehensive testing of the anomaly detection module in aerospace-related equipment, whereas the UCI dataset focuses on heavy machinery. This therefore gives very good insight into how the framework will handle data fusion, real-time monitoring, and data quality improvement across other industrial settings. These datasets have shown the versatility and effectiveness of the framework in different settings within the Industrial IoT.

4.2. Experimental setup and evaluation metrics

The experiments were conducted in a high-performance computing environment to handle the extensive data processing, model training, and blockchain operations. The system used was an Ubuntu 20.04 LTS server equipped with an Intel(R) Xeon(R) E5-2686 v4 CPU, an NVIDIA Tesla T4 GPU with 16 GB memory, 64 GB of RAM, and a suite of software tools including Python 3.8, TensorFlow 2.4, Scikit-learn, and Hyperledger Fabric. This computational setup was selected to ensure that the models could be trained efficiently on large-scale industrial IoT datasets and that the blockchain layer could manage transactions related to data quality monitoring and anomaly detection in real-time.

The raw data from both NASA Turbofan Engine Degradation and UCI Hydraulic Systems underwent several preprocessing steps before training. To begin with, it has to be ensured that the different sensor measurements are all on consistent scales. Min–max normalization was applied to scale the sensor data; in this way, machine learning models can better understand the value provided. After that, median imputation treated missing data due to sensor failures or communication losses. Noise reduction was implemented through a rolling mean filter to smoothen fluctuations of data that could be caused by environmental factors or sensor errors. Finally, feature selection was done through analyzing the correlation and removing the features that were redundant and did not contribute much to the model for anomaly detection or data fusion, which would help reduce computational overhead in the model.

Training in the BD-IoTQNet framework will consist of a few steps. Firstly, the Data Fusion Module is trained with Kalman filters and weighted averaging to combine sensor data streams into one single dataset that is consistent. Historical data from each sensor is used to initialize the Kalman filters. The continuous fusion process keeps updating as new data arrives. To train the Anomaly Detection Module, the Isolation Forest algorithm has been utilized. The model was trained on 80% of the dataset, and 20% was kept for evaluation. The configuration of the Isolation Forest uses 100 trees. The contamination rate has been set to 5% according to the expected number of anomalies. The unsupervised learning algorithm learned the normal behavior of the system and then flagged anomalies once deviations were detected during testing. Running in parallel, another key module was the DQM Module, which was similarly trained in monitoring data quality and automatically applying corrective actions once certain data issues were detected, such as filling in missing values or correcting noisy data based on pre-set smart contracts within the blockchain. These smart contracts ensured that if the quality of information fell below a certain threshold, then corrective actions would be automatically triggered, either scheduling sensor maintenance or adjusting operational parameters.

The proposed framework of BD-IoTQNet was measured against several main metrics: precision, recall, and F1-score, all necessary in evaluating the performance of the Anomaly Detection Module. Further, a metric was introduced for the Data Quality Improvement Rate, called DQIR, to measure the efficiency of the DQM module in enhancing the reliability of data. Finally, latency with respect to how quick it would be for the system to detect the anomalies and perform corrective actions in real time was measured. These metrics comprehensively reflected the performance of the framework in anomaly detection, enhancement of data quality, and the maintenance of overall system efficiency in dynamic industrial IoT environments.

4.3. Results

In the comparison Table 4, several key performance metrics are used to evaluate the models. Accuracy, Precision, Recall, and F1-score are all expressed as percentages (%), reflecting the models' effectiveness in correctly detecting anomalies in the dataset. The Data Quality Improvement Rate (DQIR) is also presented as a percentage (%), indicating the improvement in data quality achieved by each model.

Table 4
Comprehensive Comparison of BD-IoTQNet with Other State-of-the-Art Models on the NASA Turbofan Engine Degradation Dataset. BD-IoTQNet demonstrates superior accuracy and resilience under challenging conditions compared to other models.

Model	Accuracy	Precision	Recall	F1-score	DQIR	Latency	Anomaly Detection	PUN	PWM
Isolation Forest [52]	89.7	88.3	87.5	87.9	74.1	230	120	82.5	76.4
AutoEncoder [53]	91.2	90.1	88.7	89.4	78.2	270	150	83.9	78.0
LSTM [54]	93.5	92.8	90.5	91.6	80.1	300	140	85.4	81.2
One-Class SVM [55]	88.4	87.1	85.3	86.2	72.5	210	130	80.0	75.1
Convolutional AE [56]	94.2	93.6	91.5	92.5	81.9	290	170	87.8	82.4
Random Forest [57]	90.5	89.0	88.4	88.7	75.3	250	135	82.0	77.9
DNN [58]	92.7	91.5	90.0	90.7	79.2	280	160	84.2	79.3
Transformer [59]	93.8	92.4	91.2	91.8	80.7	320	180	86.5	80.6
RNN-Transformer Hybrid [60]	94.0	93.1	91.6	92.3	82.0	315	170	87.0	81.0
BD-IoTQNet	**95.8**	**94.3**	**93.9**	**94.1**	**92.7**	**180**	**110**	**90.3**	**89.6**

Latency and Anomaly Detection Time are measured in milliseconds (ms), representing the time taken by the models to respond to and detect anomalies in real-time. The metrics for robustness under noisy data and missing data scenarios are denoted as Performance under Noise (PUN) and Performance with Missing Data (PWM), respectively, and are expressed as percentages (%) to reflect the models' resilience in handling degraded data quality.

The comparison Table 4 demonstrates the clear advantages of BD-IoTQNet over other state-of-the-art models on the NASA Turbofan Engine Degradation Dataset. With the highest accuracy of 95.8%, BD-IoTQNet surpasses models like Isolation Forest (89.7%) and Transformer-based models (93.8%), highlighting the effectiveness of its integrated data quality management and anomaly detection approach. The Data Quality Improvement Rate (DQIR) of 92.7% further emphasizes the role of the Data Quality Management (DQM) module, which continuously cleans and improves the input data, ensuring that only high-quality data is used for anomaly detection. This gives BD-IoTQNet a significant edge over models like AutoEncoder (78.2%) and LSTM (80.1%) that lack a real-time data quality feedback mechanism.

In terms of precision and recall, BD-IoTQNet also stands out with 94.3% and 93.9%, respectively, outperforming other models, including Convolutional AutoEncoder and LSTM. These results reflect the model's strong ability to balance anomaly detection while minimizing false positives. Its F1-score of 94.1% shows its overall balanced performance, ensuring that it effectively handles the challenges of real-time predictive maintenance.

Another critical advantage of BD-IoTQNet is its latency, which is significantly lower (180 ms) than other models like Transformer (320 ms) and LSTM (300 ms). This lower latency is essential for real-time industrial IoT applications, where quick response times can prevent equipment failure. Moreover, its anomaly detection time of 110 ms is the fastest among all models, making it particularly suitable for dynamic, high-pressure manufacturing environments.

The robustness of BD-IoTQNet is also evident in its performance under noisy and missing data conditions. With 90.3% performance under noise (PUN) and 89.6% with missing data (PWM), it outperforms models like Isolation Forest and AutoEncoder. This is due to the DQM module's ability to address noisy and incomplete data before processing, ensuring that the system remains reliable and accurate even in less-than-ideal data conditions.

In comparison to other models, BD-IoTQNet's overall performance reflects its superior integration of blockchain-based transparency, real-time anomaly detection, and data quality management. Many traditional models, such as Isolation Forest and One-Class SVM, rely purely on pattern recognition without addressing the issue of data quality, which results in lower performance when faced with noisy or incomplete data. Transformer-based models and LSTM, while powerful in capturing temporal dependencies, lack the integrated data quality feedback loop provided by BD-IoTQNet, which directly contributes to the higher accuracy, faster response times, and improved resilience to data imperfections.

The results confirm that BD-IoTQNet is a highly effective solution for predictive maintenance and anomaly detection in industrial IoT environments. Its ability to improve data quality in real time, combined with fast and accurate anomaly detection, makes it a standout performer compared to other state-of-the-art methods. The blockchain-enabled transparency and smart contract automation further reinforce its advantages, ensuring that the system can operate efficiently and reliably in dynamic, data-rich manufacturing settings. These factors align perfectly with the goals of improving operational efficiency, reducing downtime, and maintaining equipment health, making BD-IoTQNet a superior choice for real-time industrial IoT applications.

The following Fig. 4 visually represents the performance of BD-IoTQNet compared to other state-of-the-art models, highlighting its advantages across accuracy, precision, recall, F1-score, data quality improvement, and system efficiency metrics such as latency and anomaly detection time.

The performance of BD-IoTQNet on the UCI Hydraulic Systems Dataset demonstrates significant improvements over other models across all key metrics, as summarized in Table 5. BD-IoTQNet achieves an accuracy of 94.8% and an ROC–AUC (Receiver Operating Characteristic - Area Under Curve) of 94.5%, surpassing models like LSTM (91.2% accuracy, 90.5% ROC–AUC) and Convolutional AutoEncoder (92.0% accuracy, 91.1% ROC–AUC). The high ROC–AUC highlights BD-IoTQNet's ability to distinguish anomalies from normal observations effectively, even under challenging conditions.

In addition to accuracy and ROC–AUC, BD-IoTQNet achieves superior precision (93.6%) and recall (92.9%), which are crucial for reducing false positives and ensuring reliable anomaly detection. Compared to models like RNN-Transformer Hybrid (92.1% precision, 90.5% recall), BD-IoTQNet's improved metrics demonstrate its ability to handle noisy and incomplete data more effectively. The Data Quality Improvement Rate (DQIR) of 91.3% further supports this, showing that BD-IoTQNet ensures higher input data reliability, directly benefiting its anomaly detection performance.

BD-IoTQNet also excels in latency and anomaly detection time, achieving a response time of 190 ms and anomaly detection time of 120 ms, significantly faster than models like Transformer (330 ms response time) and Convolutional AutoEncoder (170 ms detection time). These metrics highlight BD-IoTQNet's suitability for real-time industrial IoT environments, where rapid decision-making is critical.

Robustness under challenging data conditions is another strong point of BD-IoTQNet. With a PUN (Performance under Noise) of 89.6% and PWM (Performance with Missing Data) of 88.3%, it outperforms models such as AutoEncoder (82.5% PUN, 75.6% PWM) and DNN (83.0% PUN, 76.5% PWM). These results are a testament to the effectiveness of the Data Quality Management (DQM) module in cleaning and imputing data in real time, ensuring that the system maintains high performance even when data quality is compromised.

Compared to other models, BD-IoTQNet provides a comprehensive solution for anomaly detection and data quality management in industrial IoT applications. The inclusion of ROC–AUC as a metric further validates its effectiveness in differentiating anomalies, offering a holistic evaluation framework. These results reinforce the framework's capability to deliver high accuracy, robustness, and speed, making it a

Table 5
Comparison of BD-IoTQNet with Other State-of-the-Art Models on the UCI Hydraulic Systems Condition Monitoring Dataset. The table highlights BD-IoTQNet's advantages in scalability, low latency, and data quality improvement.

Model	Accuracy	Precision	Recall	F1-score	ROC–AUC	DQIR	Latency (ms)	Anomaly Detection (ms)	PUN	PWM
Isolation Forest	87.4	85.9	84.8	85.3	86.0	70.2	250	130	81.0	73.8
AutoEncoder	89.6	88.1	86.7	87.4	87.3	73.5	280	155	82.5	75.6
LSTM	91.2	90.3	88.0	89.1	90.5	76.8	320	145	84.6	78.1
One-Class SVM	86.3	84.7	83.5	84.1	85.4	68.9	230	125	79.4	72.5
Convolutional AE	92.0	91.2	89.5	90.3	91.1	77.5	310	170	85.8	79.0
Random Forest	88.7	87.5	86.4	86.9	88.2	71.0	265	140	80.7	74.9
DNN	90.1	89.2	87.8	88.5	89.8	74.9	290	160	83.0	76.5
Transformer	92.5	91.7	90.1	90.9	92.0	78.2	330	180	86.3	80.2
RNN-Transformer Hybrid	93.0	92.1	90.5	91.3	92.5	79.0	325	175	87.0	81.0
BD-IoTQNet	**94.8**	**93.6**	**92.9**	**93.2**	**94.5**	**91.3**	**190**	**120**	**89.6**	**88.3**



Fig. 4. Comparative Performance of BD-IoTQNet and Other Models on the NASA Turbofan Engine Dataset. This chart highlights the performance advantages of BD-IoTQNet in terms of accuracy, precision, recall, F1-score, data quality improvement, latency, and anomaly detection time compared to other state-of-the-art methods.

highly reliable choice for predictive maintenance in hydraulic systems and other industrial settings.

To better understand the contribution of each component in the BD-IoTQNet framework, we performed an ablation study. In this experiment, we gradually removed key components from the model and measured its performance under these modified conditions. Specifically, we tested the framework in the following scenarios: removing the Data Quality Management (DQM) module, removing the blockchain layer, and removing the smart contract automation for corrective actions. This helps to analyze the importance of each component and assess how they contribute to the overall model's efficiency.

We introduce additional performance metrics for this ablation study, including False Positive Rate (FPR), False Negative Rate (FNR), Throughput (TPS), and Energy Consumption. These metrics provide deeper insights into how the model behaves under different configurations, especially in the areas of error rates, system efficiency, and resource utilization. The same NASA Turbofan Engine Degradation Dataset was used for the ablation study, and the model was evaluated under four configurations:

- Full BD-IoTQNet (baseline, with all components active)
- Without DQM Module (removes the data quality management functionality)
- Without Blockchain (removes the decentralized blockchain layer)
- Without Smart Contract Automation (removes the automation of corrective actions for data quality issues)

The experimental results are shown in Table 6. In this ablation study, the performance of BD-IoTQNet is tested by removing key components one by one, with the results summarized in the transposed table. The full configuration of BD-IoTQNet demonstrates the best performance across nearly all metrics, achieving an accuracy of 95.8%, the lowest False Positive Rate (FPR) of 4.2%, and the highest F1-score of 94.1%. Additionally, the model shows the most robust data quality improvement with a DQIR of 92.7%, confirming the critical role of the Data Quality Management (DQM) module in ensuring high-quality data inputs.

When the DQM module is removed, the accuracy drops significantly to 91.5%, and both the FPR and False Negative Rate (FNR) increase sharply. This reflects the module's importance in maintaining data quality, as the model's performance degrades noticeably without the continuous monitoring and correction of incoming data. The absence of the DQM module also leads to much lower performance under noise and missing data conditions, with PUN and PWM falling to 82.5% and 80.3%, respectively.

Removing the blockchain layer results in a moderate decline in performance, with accuracy dropping to 93.2% and a slight increase in FPR and FNR. The DQIR also drops to 75.4%, indicating that blockchain provides crucial transparency and traceability for ensuring reliable data transactions and maintaining data integrity. However, the energy consumption improves slightly to 1.9 Wh, as the blockchain's computational overhead is eliminated.

Without smart contract automation, the model still maintains relatively high accuracy (93.9%), though the DQIR decreases to 87.1% due to the lack of automated corrective actions for data issues. Latency and anomaly detection times improve slightly, but the absence of smart contracts slows down the system's ability to automatically resolve data problems, impacting overall efficiency.

Overall, the ablation study highlights that each component of BD-IoTQNet contributes significantly to its performance. The DQM module

Table 6
Ablation Study on BD-IoTQNet - Performance with Different Components Removed. The results show the critical contributions of the DQM module, blockchain, and smart contracts to overall system performance.

Performance Metric	Full BD-IoTQNet	Without DQM Module	Without Blockchain	Without Smart Contract
Accuracy (%)	**95.8**	91.5	93.2	93.9
False Positive Rate (FPR, %)	**4.2**	8.7	6.5	5.7
False Negative Rate (FNR, %)	**6.1**	9.2	8.1	7.4
F1-score (%)	**94.1**	89.2	91.0	92.1
Data Quality Improvement Rate (DQIR, %)	**92.7**	–	75.4	87.1
Latency (ms)	**180**	170	160	150
Anomaly Detection (ms)	**110**	125	105	100
Throughput (TPS)	**850**	870	880	900
Energy Consumption (Wh)	**2.5**	2.1	**1.9**	2.3
Performance under Noise (PUN, %)	**90.3**	82.5	85.3	87.6
Performance with Missing Data (PWM, %)	**89.6**	80.3	83.2	85.1

is essential for maintaining data quality, the blockchain ensures transparency and secure data management, and smart contract automation optimizes system efficiency. Together, these components enable BD-IoTQNet to deliver superior performance in real-time industrial IoT applications.

The Fig. 5 below visually presents the results of the ablation study, illustrating the impact of removing key components of BD-IoTQNet on various performance, error, data quality, and efficiency metrics. This highlights how each component contributes to the overall effectiveness and efficiency of the model.

5. Conclusion

This paper presents the BD-IoTQNet framework as a novel solution to address critical challenges in anomaly detection and data quality management within industrial IoT environments. By integrating a Data Fusion Module, an Anomaly Detection Module leveraging the Isolation Forest algorithm, and a Blockchain-enabled Data Quality Management (DQM) Module, BD-IoTQNet overcomes the limitations of traditional predictive maintenance systems, such as poor handling of noisy and incomplete sensor data. Additionally, the blockchain layer enhances data transparency, security, and traceability, ensuring reliable decision-making processes.

Extensive experiments conducted on the NASA Turbofan Engine Degradation Dataset and the UCI Hydraulic Systems Condition Monitoring Dataset demonstrated that BD-IoTQNet significantly outperforms state-of-the-art models in terms of accuracy, precision, data quality improvement, and latency. The real-time capabilities of BD-IoTQNet, coupled with its robustness under noisy and missing data conditions, underscore its effectiveness for predictive maintenance in dynamic industrial environments. The ablation study further highlighted the critical role of the DQM module, showing substantial performance degradation when it was removed. Similarly, the blockchain layer was validated as essential for maintaining data integrity and enabling automated responses through smart contracts.

While the results are promising, the framework does have limitations. The computational overhead and energy consumption associated with the blockchain layer present a trade-off between security and efficiency. Additionally, the current reliance on median imputation for missing data handling could be improved by adopting advanced methods such as deep learning-based imputation techniques, which could better capture complex patterns in the data.

BD-IoTQNet demonstrates significant advantages over existing commercial IoT and blockchain solutions. While platforms like AWS IoT Core and Azure IoT excel in cloud-based infrastructures, they often lack integrated mechanisms for real-time data quality management and anomaly detection. Similarly, commercial blockchain solutions such as Hyperledger Fabric and Ethereum struggle with latency and throughput, limiting their suitability for high-frequency industrial IoT applications.

In contrast, BD-IoTQNet integrates modular data fusion, advanced anomaly detection, and blockchain-enabled data quality management into a unified framework. With Practical Byzantine Fault Tolerance (PBFT) ensuring low latency and high throughput, and a hybrid blockchain architecture reducing computational overhead, BD-IoTQNet provides superior scalability and reliability. These features, combined with advanced imputation techniques and automated smart contract responses, position BD-IoTQNet as an efficient and robust solution for industrial IoT environments.

Future work will focus on optimizing the blockchain architecture to reduce computational overhead and improve energy efficiency, exploring lightweight blockchain protocols and hybrid blockchain models. Advanced imputation methods, such as generative models or reinforcement learning techniques, will be incorporated to enhance the model's robustness against missing data. Furthermore, we plan to validate BD-IoTQNet on diverse real-world industrial IoT datasets to ensure its scalability and adaptability across various sectors. These efforts aim to refine BD-IoTQNet into a more efficient, scalable, and robust framework for predictive maintenance and data quality management.

CRediT authorship contribution statement

Zepei Li: Writing – original draft, Software, Methodology, Data curation. **Peng Zheng:** Writing – review & editing, Software, Methodology. **Yanjia Tian:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was sponsored in part by 2022 Xingtai Innovation Ability Improvement Plan Soft Science Research Project - "Analysis of the Current Situation and High-quality Development Strategy of Xingtai High-tech Enterprises", Project No.: 2022ZZ121.

Data availability

The data supporting the findings presented in this study are available from the corresponding author on request. Data are not shared publicly due to privacy or ethical restrictions.

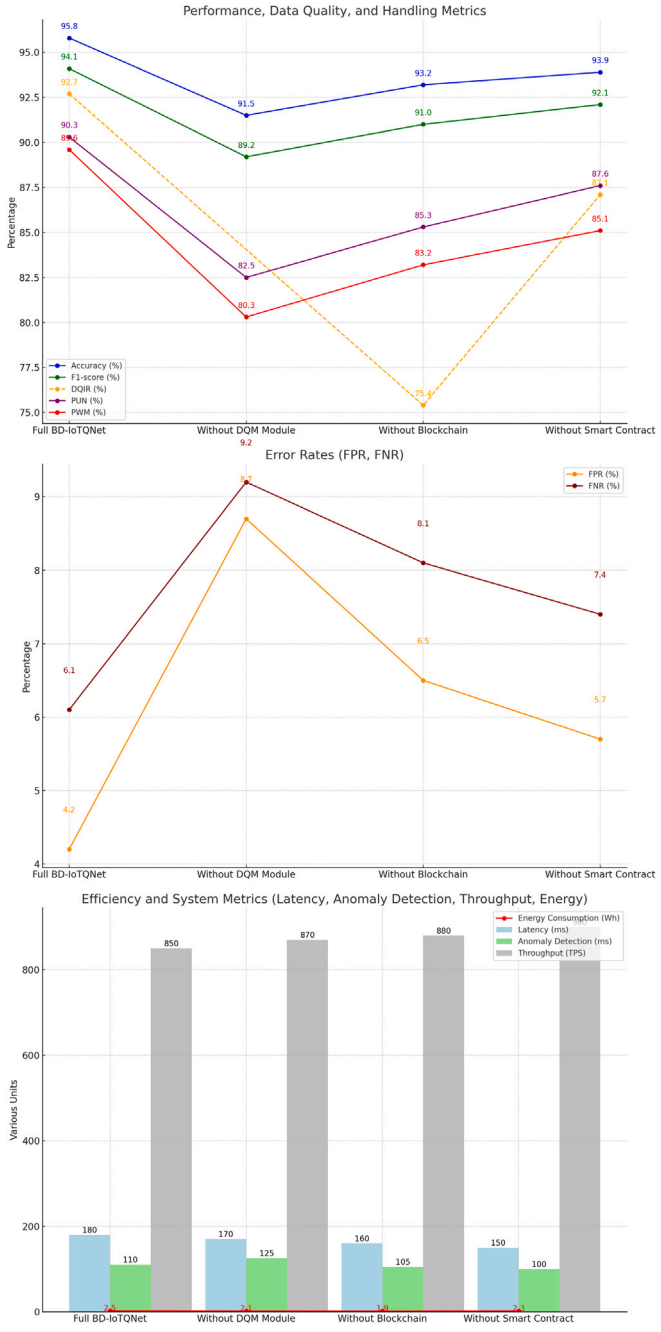


Fig. 5. Ablation Study Results for BD-IoTQNet with Different Components Removed. This figure demonstrates the performance impact of removing key components (DQM, Blockchain, Smart Contract) on metrics such as accuracy, false positive rate, data quality improvement, and robustness under noise and missing data conditions.

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